

Experiment: Family Tree Using Prolog

Aim

To implement a **Family Tree using Prolog** and determine relationships such as father, mother, grandparent, and sibling using facts and rules.

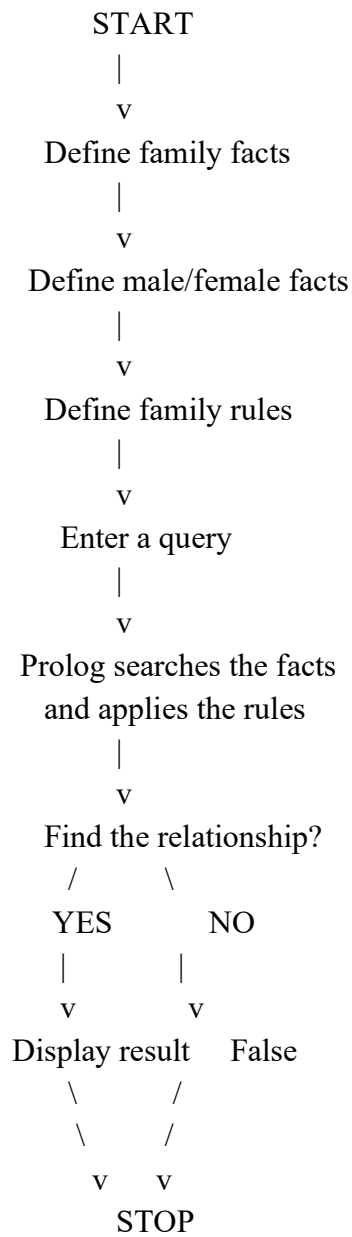
Objective

1. To represent family relationships using Prolog facts.
2. To define rules for identifying family relationships.
3. To understand the use of variables and logical inference in Prolog.
4. To execute queries and retrieve family relationships.

Algorithm

1. Start the program.
2. Define the parent relationships using Prolog facts.
3. Define male and female members.
4. Define rules for finding:
 - a. Father
 - b. Mother
 - c. Grandparent
 - d. Sibling
5. Load the program into SWI-Prolog.
6. Execute the required queries.
7. Display the results.
8. Stop.

Flowchart



Prolog Code

% Family Tree

% Parent facts

```
parent(john, mary).  
parent(john, david).  
parent(susan, mary).  
parent(susan, david).
```

```
parent(david, tom).  
parent(david, lisa).  
parent(anna, tom).  
parent(anna, lisa).
```

% Male and female facts

```
male(john).  
male(david).  
male(tom).
```

```
female(susan).  
female(mary).  
female(anna).  
female(lisa).
```

% Father rule

```
father(X, Y) :-  
    parent(X, Y),  
    male(X).
```

% Mother rule

```
mother(X, Y) :-  
    parent(X, Y),  
    female(X).
```

% Grandparent rule

```
grandparent(X, Y) :-  
    parent(X, Z),  
    parent(Z, Y).
```

```
% Sibling rule
sibling(X, Y) :-
    parent(P, X),
    parent(P, Y),
    X \= Y.
```

Sample Output

Query 1: Find the father of Mary

```
?- father(X, mary).
```

Output:

```
X = john.
```

Query 2: Find the mother of David

```
?- mother(X, david).
```

Output:

```
X = susan.
```

Query 3: Find the grandparents of Tom

```
?- grandparent(X, tom).
```

Output:

```
X = john ;
X = susan.
```

Query 4: Find the siblings of Mary

```
?- sibling(mary, X).
```

Output:

```
X = david.
```

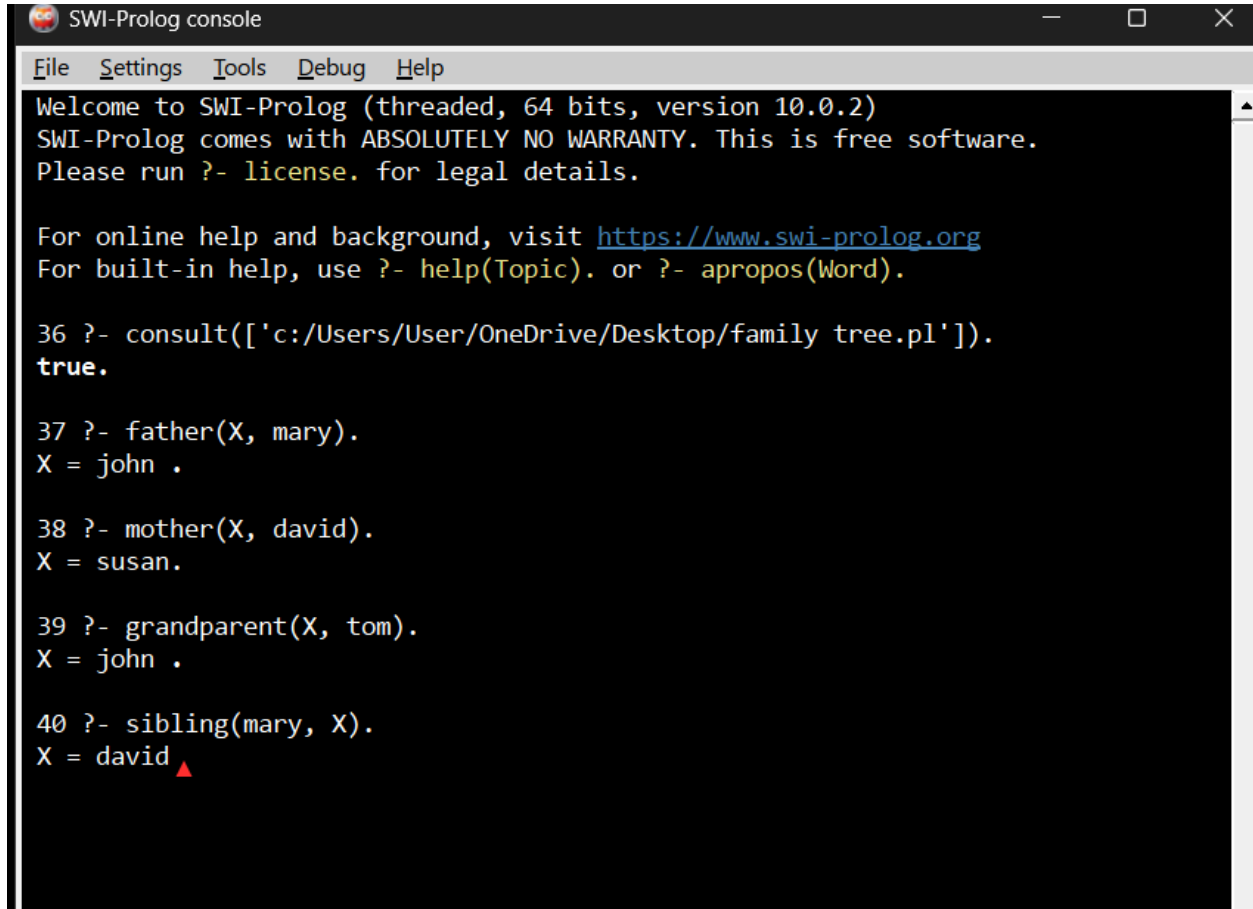
Query 5: Find the children of John

?- parent(john, X).

Output:

X = mary ;

X = david.

A screenshot of the SWI-Prolog console window. The window has a title bar with the text "SWI-Prolog console" and standard window controls (minimize, maximize, close). Below the title bar is a menu bar with "File", "Settings", "Tools", "Debug", and "Help". The main area of the window is a black terminal with white text. It shows the following text:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

36 ?- consult(['c:/Users/User/OneDrive/Desktop/family tree.pl']).
true.

37 ?- father(X, mary).
X = john .

38 ?- mother(X, david).
X = susan.

39 ?- grandparent(X, tom).
X = john .

40 ?- sibling(mary, X).
X = david ▲
```

Result

The **Family Tree** was successfully implemented in **Prolog**. The program correctly identified relationships such as father, mother, grandparent, sibling, and parent using facts and logical rules.

Conclusion

Thus, the experiment demonstrates how **Prolog can represent family relationships and perform logical reasoning**. By using facts, rules, variables, and queries, different relationships in a family tree can be determined efficiently.